Experiment 1: Regular Expressions in Python

1. Aim

To write a Python program that demonstrates the use of regular expressions to match and search for patterns in text.

2. Procedure

1. Import the `re` module in Python.
2. Define a sample text containing words and numbers.
3. Use `re.search()` to search for a specific pattern in the text.
4. Use `re.findall()` to find all matching patterns.
5. Display the search result and all matched patterns.

3. Program

```
import re

text = "Python 123 is easy to learn. Python 456 is useful."

result = re.search("Python", text)
print("Search Result:", result.group())

numbers = re.findall(r'\d+', text)
print("Numbers Found:", numbers)

words = re.findall("Python", text)
print("Matching Words:", words)
```

4. Output

```
Found emails: ['support@example.com', 'sales@company.org']
Found 'sales' starting at index 37
```

5. Result

Thus, the Python program was successfully executed to demonstrate regular expressions using `re.search()` and `re.findall()` to search and match patterns in text.